POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5014A | Semester V | Theory | 4 Credits

Sustainable Construction Practices

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5014A as Sustainable Construction Practices in Semester V for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL courses on sustainable building materials/green buildings and U.S. EPA green-building guidance. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Sustainable-material claims, material selection, structural design, building performance, green-rating submission and site implementation require current local codes, manufacturer data, tested specifications, approvals and competent professional review.

Verified source note: Verified sources support vernacular and alternative materials, construction-and-demolition waste, fly ash/slag/recycled materials, low-carbon/efficient material pathways, embodied/operational/life-cycle energy, water, VOC/IAQ, BIPV, site analysis, energy/renewables, stormwater, recycling, resilience and green-building assessment awareness. The four-module sequence is not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Sustainable Construction Practices develops a whole-life view of the built environment. It connects site-sensitive planning, resource-efficient design, low-impact materials, water and energy strategies, construction waste prevention, indoor environmental quality, resilient operation and transparent performance assessment.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Construction materials, building construction, basic environmental awareness, engineering graphics, quantity/measurement fundamentals and safe site practice.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain sustainability principles and whole-life thinking in construction.
2. Explain sustainable and alternative building materials, including resource and waste implications.
3. Explain energy, water, stormwater, indoor-environment and site strategies for responsible buildings.
4. Explain construction-stage management, green-building assessment awareness, resilience and evidence-based improvement.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain sustainability principles, life-cycle thinking and the major impacts of building decisions.	Understand
CO2	Explain conventional, vernacular, alternative and recycled-material options using transparent selection criteria.	Understand
CO3	Explain integrated site, energy, water, stormwater and indoor-environment strategies for sustainable buildings.	Understand
CO4	Explain responsible construction management, waste prevention, performance assessment and resilient operation.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Sustainability, Buildings and Life-Cycle Thinking	Sustainability in the built environment; site, materials, construction, operation and end-of-life impacts; embodied/operational/life-cycle energy; performance trade-offs and evidence.	Approx. 14 hours	CO1	Module 1
2	Sustainable Materials and Resource Selection	Conventional and vernacular materials; alternative/low-impact materials; construction-and-demolition waste; recycled/secondary resources; material-health, durability, local suitability and selection evidence.	Approx. 16 hours	CO2	Module 2
3	Integrated Site, Energy, Water and Indoor Strategies	Site analysis; passive response; energy efficiency and renewables/BIPV awareness; water conservation and stormwater; materials/VOC and indoor environmental quality; resilient landscape/service choices.	Approx. 16 hours	CO3	Module 3
4	Construction Management, Assessment and Continuous Improvement	Sustainable procurement and site practice; energy/water/waste monitoring; quality and documentation; green-building assessment awareness; handover, operation, resilience and improvement feedback.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Sustainability, Buildings and Life-Cycle Thinking

Sustainability in the built environment; site, materials, construction, operation and end-of-life impacts; embodied/operational/life-cycle energy; performance trade-offs and evidence.

CO1 · Approx. 14 hours

Sustainable Materials and Resource Selection

Conventional and vernacular materials; alternative/low-impact materials; construction-and-demolition waste; recycled/secondary resources; material-health, durability, local suitability and selection evidence.

CO2 · Approx. 16 hours

Integrated Site, Energy, Water and Indoor Strategies

Site analysis; passive response; energy efficiency and renewables/BIPV awareness; water conservation and stormwater; materials/VOC and indoor environmental quality; resilient landscape/service choices.

CO3 · Approx. 16 hours

Construction Management, Assessment and Continuous Improvement

Sustainable procurement and site practice; energy/water/waste monitoring; quality and documentation; green-building assessment awareness; handover, operation, resilience and improvement feedback.

CO4 · Approx. 14 hours

MODULE 1

Sustainability, Buildings and Life-Cycle Thinking

Sustainability in the built environment; site, materials, construction, operation and end-of-life impacts; embodied/operational/life-cycle energy; performance trade-offs and evidence.

Approx. 14 hours | CO1 | Understand · Apply

Learning goals

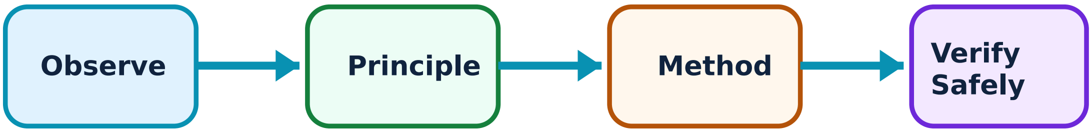
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Sustainability, Buildings and Life-Cycle Thinking: identify the system, state its principle, follow the correct method and verify the result safely.

1. Whole-life view of a building−

Building impacts arise across material extraction, transport, construction, operation, maintenance, adaptation, demolition and recovery. A sustainable decision examines resource use, emissions, water, waste, health, durability, cost and social context rather than one stage alone.

Study and application method

Create a lifecycle map for a simple building element, identifying inputs, outputs, key impact questions, information source and decisions that must be verified by a competent professional.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a lifecycle map for a simple building element, identifying inputs, outputs, key impact questions, information source and decisions that must be verified by a competent professional.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A lifecycle map is a learning tool, not a formal environmental assessment. Quantified claims need defined boundaries, data quality and recognised assessment methods.

2. Sustainability objectives and trade-offs−

Resource efficiency, climate response, water stewardship, material health, indoor environmental quality, resilience, accessibility and affordability can interact or conflict. Integrated design considers these early and tracks the assumptions behind proposed strategies.

Study and application method

Use an objective–evidence table for a design case: objective, indicator, expected benefit, trade-off, data source, responsible role and decision status.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use an objective–evidence table for a design case: objective, indicator, expected benefit, trade-off, data source, responsible role and decision status.

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Common mistake / precaution: Do not call a design “green” on the basis of one feature. Check for trade-offs such as durability, maintenance, local climate, sourcing, occupant health and end-of-life management.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Sustainable Materials and Resource Selection

Conventional and vernacular materials; alternative/low-impact materials; construction-and-demolition waste; recycled/secondary resources; material-health, durability, local suitability and selection evidence.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

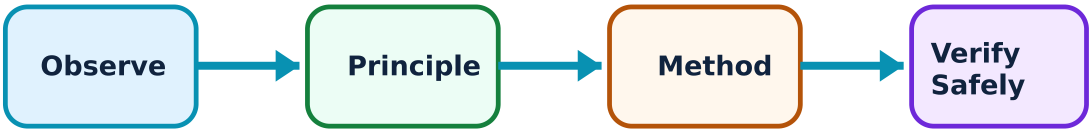
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Sustainable Materials and Resource Selection: identify the system, state its principle, follow the correct method and verify the result safely.

1. Conventional, vernacular and alternative material pathways−

Material choices may include traditional/vernacular resources, responsibly sourced conventional products and alternatives such as fly-ash/slag-based binders, recycled aggregates, laterite/quarry waste, bamboo, timber systems, aerated or geopolymer concrete and other context-appropriate innovations. Suitability depends on performance, availability, climate, skills and specification.

Study and application method

Compare two material options against required function, structural/fire/moisture performance, sourcing distance, waste content, workmanship, durability, maintenance, end-of-life and supporting test/certification evidence.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare two material options against required function, structural/fire/moisture performance, sourcing distance, waste content, workmanship, durability, maintenance, end-of-life and supporting test/certification evidence.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not substitute materials based only on environmental claims. Structural, fire, moisture, durability, health, code and supplier requirements must be confirmed for the specific project.

2. Construction waste and circular resource use−

Construction-and-demolition waste can be reduced through design coordination, accurate quantities, modularity, material protection, segregation, reuse and recovery where technically and legally appropriate. Circularity depends on traceability and safe material quality, not merely diversion statistics.

Study and application method

Prepare a waste-prevention hierarchy for a small work package: avoid, reduce, reuse, recycle/recover and authorised disposal. Identify the material stream, contamination risk, measurement method and responsible party.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a waste-prevention hierarchy for a small work package: avoid, reduce, reuse, recycle/recover and authorised disposal. Identify the material stream, contamination risk, measurement method and responsible party.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Recovered materials may have hidden damage, contamination or regulatory limits. Reuse requires inspection, documented suitability and approved applications.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Integrated Site, Energy, Water and Indoor Strategies

Site analysis; passive response; energy efficiency and renewables/BIPV awareness; water conservation and stormwater; materials/VOC and indoor environmental quality; resilient landscape/service choices.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

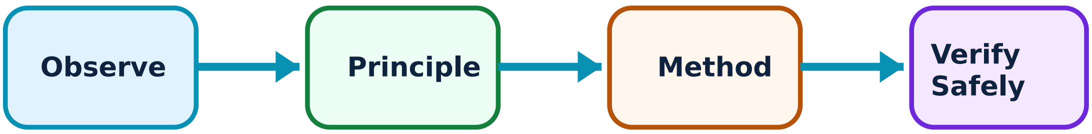
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Integrated Site, Energy, Water and Indoor Strategies: identify the system, state its principle, follow the correct method and verify the result safely.

1. Site-responsive, energy and resilience strategies−

Early site analysis informs orientation, shading, ventilation, daylight, envelope response, landscape, renewable-energy feasibility and risk resilience. Passive approaches and efficient systems can lower operational demand when matched to climate, occupancy and maintenance capability.

Study and application method

Create a site-analysis checklist covering sun, wind, rain/stormwater, vegetation, access, heat, flood/other hazards, neighbouring context and service opportunities; distinguish observed facts from assumptions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a site-analysis checklist covering sun, wind, rain/stormwater, vegetation, access, heat, flood/other hazards, neighbouring context and service opportunities; distinguish observed facts from assumptions.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Passive-strategy sketches are not final performance predictions. Thermal, structural, fire, daylight and energy design require appropriate modelling, codes and professional review.

2. Water, stormwater and indoor environmental quality−

Water conservation can combine efficient fixtures, demand management, source protection and project-appropriate reuse. Stormwater design manages runoff and site impacts. Indoor environmental quality considers ventilation, pollutants/VOCs, moisture, daylight, thermal comfort and material choices.

Study and application method

For a building case, map water inflows/outflows and indoor-environment risk points. State data required to assess potable demand, runoff, moisture, ventilation and material-emission controls.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a building case, map water inflows/outflows and indoor-environment risk points. State data required to assess potable demand, runoff, moisture, ventilation and material-emission controls.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Water reuse, drainage, stormwater and indoor-air systems must meet local public-health, plumbing, environmental and building requirements; never generalise a solution across sites.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Construction Management, Assessment and Continuous Improvement

Sustainable procurement and site practice; energy/water/waste monitoring; quality and documentation; green-building assessment awareness; handover, operation, resilience and improvement feedback.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

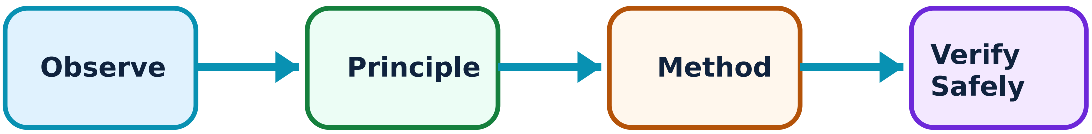
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Construction Management, Assessment and Continuous Improvement: identify the system, state its principle, follow the correct method and verify the result safely.

1. Responsible construction-stage practice−

Sustainability must be implemented on site through procurement specifications, storage/protection, erosion/sediment and pollution controls, energy/water tracking, waste segregation, quality inspections and workforce communication. Documentation connects intentions to evidence.

Study and application method

Draft a site sustainability checklist with objective, action, measure, responsible person, frequency, evidence record and escalation condition for materials, waste, water, energy and dust/noise controls.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draft a site sustainability checklist with objective, action, measure, responsible person, frequency, evidence record and escalation condition for materials, waste, water, energy and dust/noise controls.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Site controls must be compatible with approved drawings, permits, worker safety, environmental requirements and contractor method statements; do not replace formal safety/environment plans.

2. Assessment, operation and learning loops−

Green-building frameworks can organise performance targets around site, energy, water, materials, indoor environment and innovation. Actual sustainability also depends on commissioning, user guidance, maintenance, measurement and adaptation after occupancy.

Study and application method

Build a handover evidence map linking a design intention to installed system, test/commissioning record, user instruction, maintenance need, monitoring metric and review interval.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Build a handover evidence map linking a design intention to installed system, test/commissioning record, user instruction, maintenance need, monitoring metric and review interval.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A rating-framework checklist or predicted performance is not proof of actual operation. Formal certification, commissioning and performance claims require authorised processes and evidence.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Sustainability, Buildings and Life-Cycle Thinking

Use the concept flow to revise observation, principle, method and verification.

Module 2: Sustainable Materials and Resource Selection

Use the concept flow to revise observation, principle, method and verification.

Module 3: Integrated Site, Energy, Water and Indoor Strategies

Use the concept flow to revise observation, principle, method and verification.

Module 4: Construction Management, Assessment and Continuous Improvement

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Create a lifecycle-impact map for one familiar building element, identifying evidence gaps and lifecycle trade-offs.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare two material alternatives using a function–performance–resource–risk table with stated limits.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a construction waste-prevention and segregation plan for a hypothetical work package.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Complete a climate/site and water/indoor-environment checklist for a simple building brief.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop a site sustainability-monitoring and handover-evidence checklist that separates design intent from verified performance.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Sustainability, Buildings and Life-Cycle Thinking

Explain Whole-life view of a building and state one correct method, application or precaution.−

Building impacts arise across material extraction, transport, construction, operation, maintenance, adaptation, demolition and recovery. A sustainable decision examines resource use, emissions, water, waste, health, durability, cost and social context rather than one stage alone.

Answer framework: Create a lifecycle map for a simple building element, identifying inputs, outputs, key impact questions, information source and decisions that must be verified by a competent professional.

Important point: A lifecycle map is a learning tool, not a formal environmental assessment. Quantified claims need defined boundaries, data quality and recognised assessment methods.

Explain Sustainability objectives and trade-offs and state one correct method, application or precaution.−

Resource efficiency, climate response, water stewardship, material health, indoor environmental quality, resilience, accessibility and affordability can interact or conflict. Integrated design considers these early and tracks the assumptions behind proposed strategies.

Answer framework: Use an objective–evidence table for a design case: objective, indicator, expected benefit, trade-off, data source, responsible role and decision status.

Important point: Do not call a design “green” on the basis of one feature. Check for trade-offs such as durability, maintenance, local climate, sourcing, occupant health and end-of-life management.

Module 2 — Sustainable Materials and Resource Selection

Explain Conventional, vernacular and alternative material pathways and state one correct method, application or precaution.−

Material choices may include traditional/vernacular resources, responsibly sourced conventional products and alternatives such as fly-ash/slag-based binders, recycled aggregates, laterite/quarry waste, bamboo, timber systems, aerated or geopolymer concrete and other context-appropriate innovations. Suitability depends on performance, availability, climate, skills and specification.

Answer framework: Compare two material options against required function, structural/fire/moisture performance, sourcing distance, waste content, workmanship, durability, maintenance, end-of-life and supporting test/certification evidence.

Important point: Do not substitute materials based only on environmental claims. Structural, fire, moisture, durability, health, code and supplier requirements must be confirmed for the specific project.

Explain Construction waste and circular resource use and state one correct method, application or precaution.−

Construction-and-demolition waste can be reduced through design coordination, accurate quantities, modularity, material protection, segregation, reuse and recovery where technically and legally appropriate. Circularity depends on traceability and safe material quality, not merely diversion statistics.

Answer framework: Prepare a waste-prevention hierarchy for a small work package: avoid, reduce, reuse, recycle/recover and authorised disposal. Identify the material stream, contamination risk, measurement method and responsible party.

Important point: Recovered materials may have hidden damage, contamination or regulatory limits. Reuse requires inspection, documented suitability and approved applications.

Module 3 — Integrated Site, Energy, Water and Indoor Strategies

Explain Site-responsive, energy and resilience strategies and state one correct method, application or precaution.−

Early site analysis informs orientation, shading, ventilation, daylight, envelope response, landscape, renewable-energy feasibility and risk resilience. Passive approaches and efficient systems can lower operational demand when matched to climate, occupancy and maintenance capability.

Answer framework: Create a site-analysis checklist covering sun, wind, rain/stormwater, vegetation, access, heat, flood/other hazards, neighbouring context and service opportunities; distinguish observed facts from assumptions.

Important point: Passive-strategy sketches are not final performance predictions. Thermal, structural, fire, daylight and energy design require appropriate modelling, codes and professional review.

Explain Water, stormwater and indoor environmental quality and state one correct method, application or precaution.−

Water conservation can combine efficient fixtures, demand management, source protection and project-appropriate reuse. Stormwater design manages runoff and site impacts. Indoor environmental quality considers ventilation, pollutants/VOCs, moisture, daylight, thermal comfort and material choices.

Answer framework: For a building case, map water inflows/outflows and indoor-environment risk points. State data required to assess potable demand, runoff, moisture, ventilation and material-emission controls.

Important point: Water reuse, drainage, stormwater and indoor-air systems must meet local public-health, plumbing, environmental and building requirements; never generalise a solution across sites.

Module 4 — Construction Management, Assessment and Continuous Improvement

Explain Responsible construction-stage practice and state one correct method, application or precaution.−

Sustainability must be implemented on site through procurement specifications, storage/protection, erosion/sediment and pollution controls, energy/water tracking, waste segregation, quality inspections and workforce communication. Documentation connects intentions to evidence.

Answer framework: Draft a site sustainability checklist with objective, action, measure, responsible person, frequency, evidence record and escalation condition for materials, waste, water, energy and dust/noise controls.

Important point: Site controls must be compatible with approved drawings, permits, worker safety, environmental requirements and contractor method statements; do not replace formal safety/environment plans.

Explain Assessment, operation and learning loops and state one correct method, application or precaution.−

Green-building frameworks can organise performance targets around site, energy, water, materials, indoor environment and innovation. Actual sustainability also depends on commissioning, user guidance, maintenance, measurement and adaptation after occupancy.

Answer framework: Build a handover evidence map linking a design intention to installed system, test/commissioning record, user instruction, maintenance need, monitoring metric and review interval.

Important point: A rating-framework checklist or predicted performance is not proof of actual operation. Formal certification, commissioning and performance claims require authorised processes and evidence.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Sustainability, Buildings and Life-Cycle Thinking.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Sustainable Materials and Resource Selection.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Integrated Site, Energy, Water and Indoor Strategies.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.

7. State one key definition from Module 4: Construction Management, Assessment and Continuous Improvement.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Sustainability in the built environment; site, materials, construction, operation and end-of-life impacts; embodied/operational/life-cycle energy; performance trade-offs and evidence..
2. Develop a complete answer or practical plan for: Conventional and vernacular materials; alternative/low-impact materials; construction-and-demolition waste; recycled/secondary resources; material-health, durability, local suitability and selection evidence..
3. Develop a complete answer or practical plan for: Site analysis; passive response; energy efficiency and renewables/BIPV awareness; water conservation and stormwater; materials/VOC and indoor environmental quality; resilient landscape/service choices..
4. Develop a complete answer or practical plan for: Sustainable procurement and site practice; energy/water/waste monitoring; quality and documentation; green-building assessment awareness; handover, operation, resilience and improvement feedback..

LAST-MILE REVISION

Quick Revision

Module 1: Sustainability, Buildings and Life-Cycle Thinking

Sustainability in the built environment; site, materials, construction, operation and end-of-life impacts; embodied/operational/life-cycle energy; performance trade-offs and evidence.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Sustainable Materials and Resource Selection

Conventional and vernacular materials; alternative/low-impact materials; construction-and-demolition waste; recycled/secondary resources; material-health, durability, local suitability and selection evidence.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Integrated Site, Energy, Water and Indoor Strategies

Site analysis; passive response; energy efficiency and renewables/BIPV awareness; water conservation and stormwater; materials/VOC and indoor environmental quality; resilient landscape/service choices.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Construction Management, Assessment and Continuous Improvement

Sustainable procurement and site practice; energy/water/waste monitoring; quality and documentation; green-building assessment awareness; handover, operation, resilience and improvement feedback.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Whole-life view of a building	Building impacts arise across material extraction, transport, construction, operation, maintenance, adaptation, demolition and recovery.
Sustainability objectives and trade-offs	Resource efficiency, climate response, water stewardship, material health, indoor environmental quality, resilience, accessibility and affordability can interact or conflict.
Conventional, vernacular and alternative material pathways	Material choices may include traditional/vernacular resources, responsibly sourced conventional products and alternatives such as fly-ash/slag-based binders, recycled aggregates, laterite/quarry waste, bamboo, timber systems, aerated or geopolymer concrete and other context-appropriate innovations.
Construction waste and circular resource use	Construction-and-demolition waste can be reduced through design coordination, accurate quantities, modularity, material protection, segregation, reuse and recovery where technically and legally appropriate.
Site-responsive, energy and resilience strategies	Early site analysis informs orientation, shading, ventilation, daylight, envelope response, landscape, renewable-energy feasibility and risk resilience.
Water, stormwater and indoor environmental quality	Water conservation can combine efficient fixtures, demand management, source protection and project-appropriate reuse.
Responsible construction-stage practice	Sustainability must be implemented on site through procurement specifications, storage/protection, erosion/sediment and pollution controls, energy/water tracking, waste segregation, quality inspections and workforce communication.
Assessment, operation and learning loops	Green-building frameworks can organise performance targets around site, energy, water, materials, indoor environment and innovation.

LEARNING RESOURCES

References

Recommended sustainable-construction references

1. C. J. Kibert, Sustainable Construction: Green Building Design and Delivery.
2. S. V. Szokolay, Introduction to Architectural Science: The Basis of Sustainable Design.
3. Current ECBC/GRIHA/LEED and local authority guidance, as applicable to the project.
4. Current material standards, environmental product data and manufacturer documentation for specifications.

Approved public sustainable-construction sources

- https://nptel.ac.in/courses/124106455

- https://onlinecourses.nptel.ac.in/noc19_ce40/preview
- https://www.epa.gov/greeningepa/green-buildings-epa

These sources provide the verified study basis while official Course 5014A detail is unavailable; they do not replace applicable building codes, structural/environmental approvals, product certification or project-specific professional design.